

Karan Shihire

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EDUCATION

Modern College of Arts, Science and Commerce

Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0

Pune, Maharashtra

Jul 2023 – May 2026

CERTIFICATIONS

Microsoft Azure Machine Learning Fundamentals

Machine Learning Concepts, Azure ML Services, Responsible AI

Microsoft

Oracle Cloud Infrastructure Foundations Associate

Cloud Architecture, OCI Core Services, Identity & Security

Oracle

PROJECTS

• Semantic Resume Intelligence & Job Recommendation Engine

[GitHub](#)

- Built an end-to-end AI system that parses PDF resumes, extracts technical skills, and generates semantic job recommendations using transformer embeddings.
- Developed a hybrid recommendation engine combining cosine similarity and skill-overlap scoring to improve matching accuracy beyond embedding-only approaches.
- Engineered a FAISS-based Top-K retrieval pipeline to efficiently search large job datasets and implemented a retrieval → reranking architecture for final recommendation generation.
- Optimized recommendation latency by replacing exhaustive job comparisons with FAISS-based Top-K semantic retrieval.
- Designed a modular FastAPI backend with reusable services for resume parsing, embedding generation, semantic retrieval, skill analysis, and recommendation scoring.

• FaceTrack — Real-Time Attendance / Surveillance System

[GitHub](#)

- Built a real-time face recognition system capable of detecting and identifying multiple faces from live webcam streams using InsightFace and OpenCV.
- Built face enrollment pipeline handling multi-angle capture and embedding extraction, reducing false match rate during testing
 - * Challenge: matching faces reliably across lighting changes → solved using multi-angle enrollment + FAISS cosine similarity threshold tuning
- Developed embedding similarity search workflow using FAISS to perform fast identity matching on high-dimensional facial embeddings.
- Developed full-stack AI application integrating computer vision models, backend APIs, database systems, and responsive frontend interfaces.

• NetWatch — Anomaly-Based Intrusion Detection System (IDS)

[GitHub](#)

- Developed an AI-driven intrusion detection system using Isolation Forest for anomaly detection on live network traffic data.
- Built packet capture pipeline using Scapy that monitored TCP/UDP/ICMP traffic and flagged anomalies in under 100ms on local test environment
- Built a real-time Flask dashboard for visualizing packet statistics, anomaly alerts, and live monitoring data.
- Designed concurrent detection workflow for feature extraction, ML inference, and structured alert generation on real-time traffic streams.

TECHNICAL SKILLS

Languages: Python, SQL, Java

AI / ML: Computer Vision, Deep Learning, Scikit-learn, OpenCV, InsightFace, FAISS, NumPy, Pandas, Generative AI

Frameworks / Tools: FastAPI, PyTorch (Basics)

Platforms: GitHub, VS Code, Cursor, Render, Netlify, HuggingFace